

Event vs full-path vs cycle replanning for three planners

Rebuilding more often costs more and finishes fewer runs: at 20 ms checks Visibility Graph rebuilds 20.8 (event), 42.9 (full-path) and 94.6 (cycle) times per run. Across all planners, failed runs from a rebuild over 100 ms (computing time): event 36, full-path 67, cycle 73; failed runs because no route existed for 1 s (obstacles blocked every path): 23, 23, 26. Event (ours) rebuilds when the next segment is blocked; full-path also when any later segment is (Costa and Tonidandel 2024); cycle rebuilds at every check. Brackets show the change from event. In each group of three, ▲ blue (bold) marks the best value and ▼ orange the worst; planning time and time per rebuild are marked only when they differ by more than 20 %, the drift between batches. The rebuild-over-100 ms column depends on the machine; the no-route column does not.

	Runs finished (of 200)	Finished without contact (of 200)	Rebuilds per run	Rebuilds from blocked later segment (/run)	Planning time per run (ms)	Time per rebuild, median (ms)	Failed runs: rebuild over 100 ms	Failed runs: no route for 1 s	Time a route is kept (ms)
Voronoi									
Event (ours) @ 20 ms	198 ▲	95	10.8 ▲	0.2	102	4.5	2 ▲	0	605 ▲
Full-path @ 20 ms	194 (−4)	92 (−3) ▼	10.9 (+0.1)	0.1 (−0.1)	99 (−3) ▲	4.3 (−0.3) ▲	6 (+4)	0	603 (−2)
Cycle @ 20 ms	189 (−9) ▼	97 (+2) ▲	88.2 (+77.3) ▼	−	640 (+538) ▼	5.9 (+1.3) ▼	11 (+9) ▼	0	135 (−470) ▼
Event (ours) @ 100 ms	199	73 ▼	6.1 ▲	0.1	44	4.7	1	0	838
Full-path @ 100 ms	199	74 (+1)	6.1	0.0	38 (−6) ▲	4.3 (−0.3) ▲	1	0	838 ▲
Cycle @ 100 ms	199	84 (+11) ▲	16.9 (+10.8) ▼	−	117 (+73) ▼	5.8 (+1.1) ▼	1	0	497 (−341) ▼
PRM									
Event (ours) @ 20 ms	186 ▲	111 ▼	7.4 ▲	0.0 ▲	147 ▲	11.8	6 ▲	8 ▼	704 ▲
Full-path @ 20 ms	181 (−5)	115 (+4) ▲	11.6 (+4.2)	4.0 (+4.0) ▼	180 (+33)	11.1 (−0.7)	12 (+6)	7 (−1) ▲	559 (−145)
Cycle @ 20 ms	175 (−11) ▼	114 (+3)	70.4 (+63.0) ▼	−	1325 (+1178) ▼	11.7 (−0.1)	18 (+12) ▼	7 (−1) ▲	162 (−542) ▼
Event (ours) @ 100 ms	190 ▲	80 ▼	4.4 ▲	0.0 ▲	78 ▲	12.5	3 ▲	7	925 ▲
Full-path @ 100 ms	186 (−4) ▼	82 (+2)	6.4 (+2.0)	2.1 (+2.1) ▼	91 (+14)	11.2 (−1.3)	7 (+4) ▼	7	808 (−117)
Cycle @ 100 ms	188 (−2)	86 (+6) ▲	13.6 (+9.2) ▼	−	253 (+176) ▼	11.9 (−0.6)	5 (+2)	7	560 (−365) ▼
Visibility Graph									
Event (ours) @ 20 ms	180 ▲	78 ▲	20.8 ▲	0.1 ▲	411 ▲	16.3	16 ▲	4 ▲	451 ▲
Full-path @ 20 ms	168 (−12)	72 (−6) ▼	42.9 (+22.1)	23.7 (+23.6) ▼	732 (+320)	14.1 (−2.2)	27 (+11) ▼	5 (+1)	257 (−194)
Cycle @ 20 ms	165 (−15) ▼	72 (−6) ▼	94.6 (+73.8) ▼	−	1686 (+1274) ▼	14.8 (−1.5)	27 (+11) ▼	8 (+4) ▼	129 (−322) ▼
Event (ours) @ 100 ms	188 ▲	61 ▼	8.1 ▲	0.0 ▲	157 ▲	16.6	8 ▲	4	787 ▲
Full-path @ 100 ms	182 (−6) ▼	64 (+3)	12.7 (+4.6)	4.8 (+4.8) ▼	221 (+63)	14.4 (−2.2)	14 (+6) ▼	4	619 (−168)
Cycle @ 100 ms	185 (−3)	67 (+6) ▲	18.5 (+10.4) ▼	−	350 (+193) ▼	15.1 (−1.5)	11 (+3)	4	491 (−296) ▼